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Quantitative Molecular Imaging using Deep Magnetic Resonance Fingerprinting

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Method Article

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Abstract

Deep learning-based saturation-transfer magnetic resonance fingerprinting (MRF) is an emerging approach for noninvasive *in vivo* imaging of proteins, metabolites, and pH. It involves a series of steps, including (i) sample/subject preparation, (ii) image acquisition schedule design, (iii) biophysical model formulation, (iv) artificial intelligence (AI) and computational model training, followed by (v) image acquisition, (vi) deep reconstruction, and (vii) analysis. The saturation-transfer-based molecular MRI field has been slow to reach clinical maturity and adoption for clinical practice due to its technical complexity, semi-quantitative contrast-weighted nature, and impractically long scan times needed for the extraction of quantitative molecular biomarkers. Deep MRF constitutes a remedy for these challenges, providing a quantitative and rapid framework for extracting biologically and/or clinically meaningful molecular information. Here, we define a standardized protocol and a practical, complete guide for quantitative molecular MRI using deep MRF. We describe *in-vitro* sample preparation, animal and human scan considerations, and provide intuition behind acquisition protocol design and optimization of chemical exchange saturation transfer (CEST) and semisolid magnetization transfer (MT) quantitative imaging. We then extensively describe the building blocks for several AI models and demonstrate their performance for different applications, including cancer monitoring, brain myelin imaging, and pH quantification. Finally, explicit guidelines for further modification and expansion of the pipeline for imaging a variety of other pathologies (such as neurodegeneration, stroke, and cardiac disease) are given, accompanied by detailed open-source code and sample data. The protocol takes ~48 min - 57 hrs to complete by a scientist at the graduate student level.

Introduction

Development of the protocol

The sensitivity of chemical exchange saturation transfer (CEST) magnetic resonance imaging (MRI) to pH and protein and metabolite concentrations is proving to be a powerful tool for molecular imaging of a wide range of pathologies[1,2,3]. CEST MRI uses frequency-selective radio-frequency (RF) pulses to saturate the magnetization of exchangeable protons on a variety of molecules, including mobile proteins and metabolites. Fast chemical exchange with bulk water protons then results in CEST contrast via a decreased water MRI signal. The exchangeable protons of different metabolites and proteins have different resonant frequencies and different chemical exchange rates providing selectivity for detecting the molecule of interest. The CEST contrast depends on the chemical exchange rate (k_{ex}), which is pH sensitive, and on the volume fraction of the exchangeable proton pool (f_s), which depends on the protein and metabolite concentrations. Since innumerable pathologies are characterized by early changes in pH, protein synthesis, and/or metabolism, CEST can play a vital role in the assessment of disease progression and individual response to therapy. For example, the amide proton CEST contrast from endogenous proteins has been used to detect changes in pH during stroke that provide insight into the viability of the ischemic penumbra[4,5,6,7,8] and to distinguish tumor progression from radiation necrosis in clinical glioma patients[9,10,11]. Similarly, amine exchangeable protons of glutamate have been used to measure changes in brain glutamate concentrations in neurodegenerative diseases[12,13,14] and brain tumors[15], and guanidyl exchangeable protons of creatine have been used to measure creatine concentrations in the heart that may be predictive of heart failure[16,17]. In addition, CEST reporter genes[18,19,20] have been used to image biological therapeutics such as oncolytic virotherapy[21] and adeno-associated viral (AAV) gene transfer for cardiac gene therapy[22].

MR spectroscopy (MRS) has previously been the main tool used for *in vivo* metabolite imaging. However, MRS suffers from several challenges. First, due to its inherently low sensitivity, the spatial resolution is typically very limited, and therefore, small pathologies or treatment effects can easily be missed due to partial volume effects. Furthermore, at clinical field strengths (≤3T), many of the metabolite resonances overlap (e.g., glutamate and glutamine), complicating the spectral interpretation and metabolite quantification. In contrast, in CEST MRI experiments, the accumulation of the exchangeable protons in the free water pool by long saturation labeling, together with the fast chemical exchange rate of

CEST compounds, acts as a signal amplifier and allows to achieve 1-3 orders of magnitude increased sensitivity compared to MRS methods. In addition, CEST can probe not only certain metabolites but is also sensitive to protein concentration and tissue pH, providing further molecular insights.

However, clinical translation of these CEST methods has been hindered by the semi-quantitative nature of the image contrast and the difficulty disentangling the pH and protein/metabolite concentration contributions to the CEST contrast. This is particularly problematic for imaging tumors where slightly increased intracellular pH and decreased extracellular pH are typically observed[23,24,25,26,27] and can change with tumor progression or in response to tumor therapies[28,29]. Efficient and selective methods for the quantification of proton k_{ex} and f_s are needed to produce high quality pH and protein/metabolite concentration maps and help move these studies forward into the clinic.

A rapid Magnetic Resonance Fingerprinting (MRF) based CEST technique was recently pioneered that enables accurate quantification of proton exchange rates and volume fractions[30,31,32]. Much like conventional MRF methods[33,34], CEST-MRF varies the image acquisition parameters pseudo-randomly to generate unique signal trajectories for different tissue parameters. MRF works in the transient state where each image acquisition in the acquisition schedule depends on the prior history of acquisitions so that the signal patterns for different tissue parameters evolve differently and are distinct. The measured signal trajectories are matched to a large dictionary of signal trajectories computed using the Bloch equations for different tissue parameter combinations. The CEST-MRF signal depends on k_{ex} and f_s as well as the water proton pool T_1 and T_2 relaxation times, all of which can be mapped with MRF and provide clinically useful data with acquisition times of only a few minutes.

Dictionary-based methods for MRF parameter map reconstruction are typically limited to matching only a few parameters, since the size of the dictionary grows exponentially with the number of parameters (making the matching of highly multi-parametric data computationally expensive). Even for the reconstruction of a few tissue parameters, dictionary matching can take many hours. In addition, the accuracy of the parameter maps is limited by the resolution of the matching dictionary, which is simulated for discrete parameter values only. The application of MRF methods for many disease pathologies, in which multiple tissue parameters are changing with time, is therefore very challenging, as a wide range of values for multiple tissue parameters will need to be included in the matching dictionary. Large dictionaries result in long reconstruction times, but also require significant computational resources to generate in the first place. Dictionary compression methods[35,36] allow a reduced reconstruction time but still require a full dictionary to be generated. In contrast, low-rank approaches[37] eliminate the need for full dictionary generation, but at the cost of increased computation time and a reliance on polynomial approximations that can reduce the accuracy of the reconstruction. To overcome these challenges, several recent studies have implemented deep learning neural network based MRF reconstruction methods that train a neural network with sparse dictionaries[29,38,39,40,41,42,43,44]. The neural network approach has the advantage of being both more efficient and accurate as it generates continuous tissue parameter values instead of being limited to discrete dictionary values.

Here, we define a standardized protocol for saturation transfer (CEST and semisolid MT) deep MRF. We provide a detailed guide for all necessary and optional steps, including in-vitro sample preparation, animal and human imaging considerations, acquisition details, and AI-based reconstruction. A detailed open-source code and sample data for all framework steps are available at https://github.com/momentum-laboratory/molecular-mrf.

Applications

Initial applications of CEST-MRF have been demonstrated in both preclinical and clinical studies of brain tumors. A preclinical study of oncolytic virotherapy in a mouse glioma model observed significantly lower amide proton concentration and exchange rate in the viral infected core of the tumor compared to tumor rim and contralateral regions[29]. Both effects are indicative of cell apoptosis as it is known to both inhibit protein synthesis[45] and decrease cytosolic pH[28]. The area of decreased amide exchange rate increased with time as the virus spread throughout the tumor and colocalized with caspase-3 staining of the apoptotic tumor region in histology[29]. In contrast, no significant differences between apoptotic and non-apoptotic tumor regions of interest (ROIs) were observed in conventional MRI and CEST biomarkers (T_1, T_2, MTR_{asym}). A recent clinical CEST-MRF study of brain metastases observed significant differences in amide proton volume fraction and exchange rate between edema, solid tumor, necrotic tumor, and contralateral brain tissue ROIs[41]. Furthermore, brain metastases from different primary tumors showed varying exchange parameter values. These studies demonstrate the great potential of CEST-MRF to detect very early signs of therapeutic efficacy and tumor apoptosis and for distinguishing different tumor growth phenotypes.

While these initial CEST-MRF studies focused on cancer-related applications, saturation transfer MRF could be applied to a much broader range of pathologies. For example, several recent works were focused on molecular imaging of the leg muscle[42,46]. Recent in-vitro results have further demonstrated the promise of CEST MRF for expansion into a variety of other molecular targets and applications[47]. Future in-vivo CEST-MRF targets may include iodine-based contrast agents for the quantification of extracellular pH and dual modal CT/MRI[48] amine protons for the quantification of glutamate in neurological disorders[12], guanidyl protons for the quantification of creatine in cardiovascular disease[13], and aliphatic protons for the quantification of lipids and protein folding[49] in Alzheimer's disease[50], liver disease[51], and cancer[52]. To enable researchers to further expand the horizons of saturation transfer MRF, either by pursuing new metabolic targets or through the development of new AI approaches, this protocol will provide a detailed description and code for *all* pipeline stages and illustrate the intuition behind each methodological setup.

Expertise needed to implement the protocol

The protocol was designed and organized to be successfully implemented by a scientist at the graduate student, or higher, level. It requires basic knowledge of Python, basic experience with machine learning libraries (PyTorch), and access to preclinical or clinical MRI scanners.

Overview of the procedure - key components and considerations

The key steps of the deep molecular MRF pipeline are illustrated in **Fig. 1**. In the first (pre-experiment) stage, the biological scenario of interest needs to be determined and characterized, including the pathology and species to be investigated (e.g., mice or humans), and associated molecular environment (the metabolites involved, and the general magnetic tissue properties expected). According to these broadly defined characteristics, digital phantoms and tissue mimicking in-vitro models are assembled, and animal/human cohort properties are determined (**Fig. 1**, **left**). Next, acquisition protocols are designed to accommodate sufficient parameter discrimination ability and are organized as a dual computer/scanner instruction file.

In the second (computational/AI) stage, the biophysical model associated with the experiment is computationally realized, via explicit analytical solution or a numerical Bloch-McConnell equation solver. Next, a dictionary of synthetic MRI signals is simulated and used for training a deep reconstruction neural network (**Fig. 1**, **center**).

In the last (biological/clinical application) stage, the MRF acquisition instruction file is loaded into the MRI scanner and used for rapidly acquiring the raw, molecular information encoding, images. Finally, the reconstruction network inference takes place, and the raw image input is translated into quantitative parameter maps (**Fig. 1**, **right**).

Notably, if the molecular target of choice was already imaged using CEST MRF for the same biological/clinical application and documented in the literature, several steps can be skipped, including: acquisition protocol design and optimization, biophysical model design, and neural network training. In this case, previously reported acquisition protocols (see **Table 1**) and trained neural networks (provided in https://github.com/momentum-laboratory/molecular-mrf) can be directly employed.

Sample preparation and pre-experimental steps

Choosing a saturation transfer compatible biological question

Saturation transfer MRI is attractive for providing biophysical information in a wide range of biological cases, including glutamate imaging in neurological disorders[12], mobile protein and pH characterization in cancer and ischemia[11], muscle creatine metabolism quantification[53], and more. However, several conditions must be fulfilled for a biological question to become relevant for this contrast mechanism:

1. The target imaging compound must contain exchangeable protons.

2. The concentration of the target compound generates sufficient signal-to-noise ratio (SNR). This condition is typically satisfied in the millimolar concentration range.

3. The proton exchange rate (k_{ex}) is in the slow to intermediate regime, namely smaller than the chemical shift ($\Delta\omega$) in radians/sec[54].

Detailed tables containing the proton exchange rate, chemical shift, number of labile protons, and in-vivo concentrations for a variety of compounds and metabolites are available at[2,55,56].

Sample preparation and subject cohort design

Once a saturation-transfer-compatible biological question is defined and characterized, a series of experiments under different biological conditions is recommended to establish and validate the imaging performance of the MRF pipeline (**Fig. 2**).

First, numerical simulations and digital phantoms can be generated using a BM-simulator. Next, physical samples can be used, starting from in-vitro solutions of the compound of interest. A number of test tubes (either NMR, Falcon tubes, or others)[57,58] are assembled, containing different compound concentrations at various pH levels. While the first phantom study (aimed for acquisition protocol development) can have a small number of vials, bearing the average, minimum, and maximum concentration and pH levels expected for the target in-vivo environment, a second round of phantom experiments should later be conducted with a finer resolution of concentration and pH levels. This experiment should also mimic a physiological multi-metabolite background environment, e.g., using bovine serum albumin (BSA), which generates amide, amine, and relayed nuclear Overhauser effect (rNOE) signals, cross-linked BSA or hair conditioner (for generating a semisolid MT contribution), gadolinium, and $MnCl_2$ (for adjusting the water relaxation times). Another important consideration is the phantom temperature, which affects the proton exchange rate. It should ideally be set to physiological 37°C and maintained using hot air blowers, circulating hot water pumps, or temperature-maintaining pads.

Temperature monitoring can be performed using MRI-compatible sensors (positioned as close as possible to the vials, or inserted into them, if possible).

Once a successful phantom validation is reached (see additional details in the ‘Image analysis, interpretation, and validation’ section below), an in-vivo animal study can be designed, typically involving disease model mice/rats and wild type matched controls. Unlike human studies, rodent imaging is commonly performed using dedicated high field preclinical scanners and continuous wave (CW) saturation pulses. Therefore, a rigorous modification of the acquisition protocol must take place before scanning humans. If the in-vitro step was not performed on clinical scanners, it should be repeated and re-confirmed using the modified schedules. Next, a healthy volunteer study can take place (after obtaining ethical approvals and following institutional and national guidelines). Following confirmation that all previous steps yielded sufficient agreement with the ground truth, literature values, and reference non CEST-MRF imaging metrics, a pilot study on clinical subjects may take place, where a patient population and matched healthy controls are imaged, and the resulting quantitative CEST-MRF maps are validated against gadolinium enhanced images, MRS, diffusion-perfusion images, PET images, tissue biopsies (if available), and ultimate clinical outcome (determined by the physician).

Choosing the static MRI field (B_0) strength

The MRI scanner static field needs to be carefully chosen for a given CEST target. For example, according to the slow to intermediate exchange rate condition (mentioned in the previous sections), imaging the amine protons at 3 ppm, assuming an exchange rate of k_{ex} = 5500 s^{-1} [12] is feasible at 7T ($\Delta\omega$ = 7T x 42.57 (MHz/T) x 2π rad x 3 ppm ≈ 5614 rad/s) but not at 1.5T[59]. Generally, increased B_0 field translates into longer longitudinal relaxation times, narrower peak widths, and reduced signal spillover between peaks, which are favorable for CEST SNR and multi-pool signal separability[60]. On the other hand, high-field scanners are expensive, less accessible for human studies[61], characterized by a higher degree of B_0 and B_1 field inhomogeneity and increased specific absorption rate (SAR), which imposes limits on the saturation pulse powers, durations, and duty cycle.

Two practical means may assist in determining the optimal field strength prior to the experiment. (1) Building on previously used field strengths, as described in CEST literature[3,55,62]. (2) Simulating the expected CEST signals and their associated SNR across various compound concentration and exchange rate, for a variety of field strengths.

Acquisition schedule design

While CEST-MRF methods can provide a powerful tool for quantitative molecular imaging, one limitation is the need to tailor the image acquisition schedule for the clinical case of interest. The initially developed MRF methods have employed pseudo-random acquisition schedules; however, recent studies[47,63,64,65,66,67] have demonstrated that the accuracy and efficiency of MRF sequences can be significantly improved by optimal design of the acquisition schedule. Optimization not only provides improved accuracy and discrimination of the tissue parameters but also allows for a significantly reduced number of MRF acquisitions, further reducing the image acquisition time.

A typical CEST-MRF acquisition protocol is shown in **Fig. 3**. It comprises a frequency selective pre-saturation pulse, a rapid readout module (e.g., single-shot EPI[29,30], 3D turbo spin-echo[68,69], or 3D snap-shot EPI/GRE[42,70,71], concluded by a recovery period. While CEST-weighted optimization aims for increased CEST SNR, high contrast-to-noise (CNR) ratio, and mitigated RF spillover[72], quantitative CEST-MRF protocol design is focused on parameter discrimination ability, quantification accuracy, and minimized acquisition time. However, unlike water T_1/T_2 MRF, where the raw images can be acquired using extremely short TRs (e.g. <15 ms[33]), each CEST-MRF raw image must be acquired using a much longer repetition time due to the low SNR of non-steady-state CEST signals. The SNR requirement necessitates amplification using a 1-5 s pre-saturation pulse (or pulse train)[30,73] and a recovery time of ~1 s or longer[47]. Similar to classical CEST quantification studies[74], the saturation pulse power range used is crucial for discriminating different exchange parameters and is generally chosen to include a wide enough range spanning the optimal labeling efficiency ($\alpha \approx 1$[75]) and lower.

Several computational approaches can be implemented for obtaining an optimized CEST-MRF pulse sequence. First, a sparse search space or a grid of relevant acquisition parameters needs to be defined, e.g., saturation pulse power between 0 to 6 uT with 0.1 uT increments, saturation pulse duration of 1 s to 3 s with 0.1 s increments, etc. For each acquisition parameter combination, a simulated dictionary can then be generated and followed by an estimation of the discrimination ability using: (i) the off-diagonal Frobenius norm dot product loss, which estimates the correlation between various dictionary entries[73]; (ii) a Monte Carlo simulation of noise propagation, where white Gaussian noise is added to the dictionary and the resulting trajectories are matched to the original noiseless dictionary[73]; or the (iii) Cramer Rao lower bound[76]. A sample code implementing for each of these metrics is available in the folder "metric_example" of the provided repository https://github.com/momentum-laboratory/molecular-mrf. However, given the large number of tissue exchange properties being matched and the large number of experimental parameters that can be varied (saturation pulse power, time, and frequency offset as well as the excitation flip angle and repetition time) optimization of the acquisition schedule is greatly complicated by the vast optimization search space. Recent advances in machine learning optimization methods have helped overcome some of these challenges and demonstrated improved accuracy of the reconstructed chemical exchange parameter maps while at the same time enabling a significant shortening of the acquisition time[47,67,77]. Given their complexity compared to the previously mentioned methods, they are beyond the scope of this protocol, which aims to enable a graduate-student-level scientist to successfully reproduce molecular MRF. To benefit from these AI-based-optimization studies, however, the detailed acquisition protocols for various molecular targets generated using these AI optimization methods is provided in the folder supplementary/published_pulse_sequences, in the provided GitHub repository (https://github.com/momentum-laboratory/molecular-mrf/), as described in **Table 1**. An optional code demonstrating the AI-based optimization pipeline (for advanced users) is available at https://github.com/Heo-Group/LOAS-MRF.

Computational modeling and AI-based reconstruction

Biophysical model formulation

A crucial pillar for MRF reconstruction is a well-formulated biophysical model, which closely mimics the biological scenario of interest. For semisolid MT and CEST MRF, the Bloch-McConnell (BM) equations constitute such a thoroughly explored and validated model[78], yet several key parameters must be explicitly set or constrained for practical use. Two types of information are needed at this step: tissue magnetic properties and acquisition schedule parameters. The tissue properties are derived from the biological question, compound target, and the species to be imaged (determined in the previous sections). These specifications translate into the range of biologically feasible T_1 and T_2 relaxation times (for both water, CEST and semisolid MT pools), the number of participating proton pools, the solute concentrations, chemical shifts, and proton exchange rates. All these parameters can be derived from the CEST literature (e.g., see the compound property tables at[2,55,56]). The acquisition schedule parameters include the saturation pulse powers, durations, frequencies, RF-shapes, duty cycle, and delays, together with the gradient, spoiler, slew-rates, and readout specifications. All these parameters can be designed and optimized as described in the previous section. Unlike the tissue parameters, which are given as a broad range (to accommodate both physiological and pathological cases), the acquisition parameters need to be defined as accurately as possible. In this protocol, we organize the biological tissue properties and acquisition protocol parameters into human and machine readable .yaml and .seq files, respectively. These two files serve as input for a BM equation solver, which simulates the expected MR signals for each tissue parameter combination. The resulting data eventually constitutes an extensive dictionary of synthetic saturation transfer signals and their paired ground truth quantitative parameters.

Dictionary simulation and dot-product matching

In the context of MRF, a dictionary refers to a library of signals which represent the expected scanner output for a given acquisition schedule and a certain combination of tissue parameters. The dictionary is generated by solving the Bloch-McConnell equations, yielding a step-wise representation of the magnetization vector at the end of each RF/delay event, concluded by the read-out associated signal intensity. Although an analytical solution exists for some cases, it does not hold or is insufficiently accurate for a non-steady-state saturation pulse train (required by most clinical scanners). Notably, for some MRF scenarios (e.g., multi-pool cancer imaging), the number of combinations can reach into thousands or even millions of entries, necessitating a time-efficient implementation.

The dictionary simulator provided in this protocol comprises three main components: the Pulseq sequence definition[79], a C++ simulator backend, and a Python front-end with parallelization capabilities. The C++ backend mirrors the one used in Pulseq-cest[80], enhanced with a C++ interface for MRF simulation to facilitate simulation parameter updates. It processes .seq files pulse-by-pulse, extracting pulse information, and updating the simulation matrix accordingly. Next, the Bloch-McConnell equations solution is iteratively computed[81,82] using the Pade approximation for matrix exponentiation[83]. A SWIG-generated wrapper allows the C++ code to be utilized as an external library in Python. The Python front-end, designed for MRF simulations, leverages several BMCTool library definitions to initialize simulation parameters and dynamically updates them for each parameter set, while managing C++ code calling and employing parallelization for increased efficiency.

Once the dictionary is obtained, it can be used as a direct comparison means for the signals acquired in real-world experiments and de-facto solve the parameter quantification inverse problem. Signal matching can be performed using several algorithms, with the simplest being the vector dot product. In this protocol, the matching was performed straightforwardly using the Python NumPy library and executed in batches to reduce RAM consumption.

Supervised deep reconstruction of quantitative saturation transfer maps

The computational complexity of dot-product-based quantitative image reconstruction is directly affected by the size of the dictionary. While such reconstruction is feasible in two pool imaging cases (such as in-vitro phantoms), in-vivo imaging typically involves 3 proton pools or more, where the image reconstruction time increases exponentially with each added tissue property. The resulting dictionaries typically contain millions of entries, which translates into very long reconstruction times (e.g., more than two hours for a single slice image[29]). Another limitation associated with dot-product matching is the discrete nature of the dictionary. Deep reconstruction neural networks constitute an efficient means to overcome these two challenges. Previous studies have shown that a neural network, trained using purely simulated signal trajectories, can successfully learn the continuous mapping between raw, MRF information encoding data, and decoded quantitative exchange parameters[29,38,39]. In practice, the synthetic dictionaries are normalized (using an unsaturated M_0 image, 2-norm normalization, or zero-mean unit-variance signal whitening), injected with white Gaussian/Rician noise to promote learning robustness, and are then fed as 1D tensors into a fully connected neural network. At inference time, the temporal dynamics of each pixel in the acquired data constitute a 1D tensor, which undergoes the same normalization and serves as input to the trained network.

Sequential learning of multi-metabolite information

For complex pathologies the relaxation and exchange parameters of several molecular pools may vary simultaneously, posing an additional challenge for the discrimination ability of a given CEST-MRF protocol. An optional step for addressing this issue is the serial acquisition of several MRF protocols, where each is designed to isolate a particular pool of interest. For example, a first MRF protocol may vary the flip angle and repetition times without any off-resonance pulses to quantify the water T_1 and T_2. A second protocol may vary the saturation pulse frequency offset and power, while using large frequency offsets (e.g., > 8 ppm), to isolate the effects stemming from the semisolid MT, and a third protocol could use saturation pulse frequency offsets aimed at the chemical shift of the compound of interest (e.g., 3.5 ppm). To decode the acquired data, a series of neural networks (**Fig. 4**) can be trained using synthetic signal dictionaries for each acquisition protocol, where the output of each NN is fed as input for the next. Such reconstruction architecture serves as a practical means to sequentially reduce the parameter space size and may improve the reconstruction performance[29].

Unsupervised learning

Supervised learning-based MRF reconstruction approaches have demonstrated the ability to efficiently learn a mapping relationship between a high-dimensional MRF space and a low-dimensional tissue parameter space, bypassing the exhaustive dictionary search and circumventing the curse of dimensionality. Nevertheless, to improve the reconstruction accuracy and generalization performance, supervised learning methods still require a very large training dictionary consisting of all possible tissue parameter combinations, which is extremely computationally expensive. An alternative approach is to use a label-free unsupervised learning algorithm to derive unique MRF features directly from *in-vivo* data, ideally reflecting realistic (rather than simulated) tissue properties. In a previous study[40], a convolutional neural network (CNN)-based, deep-learning architecture was designed to solve the ill-posed MRF inverse problem in an unsupervised fashion (**Fig. 5**). Rather than minimizing the difference between output predictions and given ground-truth labels on a training dataset, the neural network learned to synthesize the given input by solving an MRI physics (Bloch-McConnell equations)-based forward model. Applying convolutional kernels on the input MRF images can take advantage of additional information by aggregating the feature vectors of neighboring pixels in the spatial dimension, thus providing further de-noising capabilities.

Data acquisition

Once the acquisition schedules are designed and the imaging sample (or subject) is ready, data acquisition may take place. While traditional CEST-weighted and water T_1/T_2 MRF imaging are readily available for some scanners and vendors, there is no off-shelf commercial product for saturation transfer MRF. To address this gap and foster the utilization and future development of molecular MRF, we provide two open-source pulse sequence packages:

(1) A preclinical Bruker CEST-MRF pulse sequence compatible with ParaVision 6 and Paravision 360 platforms. It enables varying the repetition time, flip angle, and the saturation pulse power, duration, and frequency offset for the various images acquired, and the use of either EPI or RARE image readouts. The complete pulse sequences can be downloaded from https://osf.io/52bsg.

(2) A vendor-agnostic hardware independent CEST-MRF pulse sequence implemented using the pulseq prototyping framework[79,80]. It can be used on various clinical and preclinical scanner platforms and enables reproducing the exact definitions of the CEST preparation period used in previous reports (**Table 1**)[29,30,40,41,42,67,73,77]. The readout pattern needs to be separately determined via the pulseq environment (which can be openly programmed in Python) or combined with vendor-specific readout modules/scripts (as performed in[80] (yet this option requires a dedicated user agreement (C2P)). Another attractive element in the pulseq-based implementation is that the same pulse sequence file may be directly employed for *in-silico* simulation.

In both packages, a single file (.txt or .seq format) that contains the acquisition parameters needs to be loaded into the scanner prior to the actual scan (e.g., via ssh).

In addition to the core CEST MRF pulse sequences, additional acquisition of $T_1/T_2/B_0/B_1$ maps may be required (if the sequential learning approach is being implemented, see above). This can be performed using vendor-specific widely available pulse sequences or using pulseq. Notably, conventional CEST-weighted imaging can be acquired (for comparison with CEST-MRF findings) by applying minor modifications to the MRF pulse sequence text files.

Image analysis, interpretation, and validation challenges

Once acquired, the raw CEST-MRF data needs to undergo standard pre-processing steps, which include skull-stripping (for brain imaging), background removal (which can be achieved by setting a threshold on the dot product map, by using a human/mouse atlas, or manually) and motion correction (if needed). As saturation transfer MRF relies on comparing simulated signals to experimentally acquired data, it is essential to normalize both in the same manner. This can be achieved by dividing all signals by an M_0 image, by using 2-norm normalization, or by using zero-mean unit-variance signal whitening.

Next, the reconstruction of quantitative exchange parameter maps takes place. While this can be performed by dot-product matching of the acquired data to the entire simulated dictionary, it is much faster and recommended to employ neural networks (trained as described in the previous sections) for rapid inference.

Several layers of validation can then be applied, depending on the imaged specimen (**Fig. 2**). For synthetically simulated images, various levels of noise can be injected into the raw signals prior to the reconstruction, followed by assessment of the reconstruction accuracy (exploiting the in-silico model-based proton volume fraction and exchange rates as ground truth).

For in-vitro phantoms, the concentration maps can be compared to the concentrations calculated from the known molecular weight and the measured compound mass and solution volume. The proton exchange rates can be compared to QUESP measured exchange rates or pH meter values (in case the translation between proton exchange rate and pH is known in the literature).

For *in-vivo* animal imaging, the ability to detect spatially dependent molecular changes using CEST MRF can be examined by comparison to histology and immunohistochemistry (IHC). If available, additional imaging methods such as ^{31}P MRS, ^{1}H MRS, and PET can serve as validation tools for intracellular pH and compound/metabolite content.

Human healthy volunteers, represent an important translational ability milestone, where the CEST-MRF derived quantitative exchange parameters of the target tissues (e.g, white matter, gray matter, leg muscle, cardiac muscle, etc.) should be compared to known literature values[12,81,85,86,87,88,89,90,91]. This step can be automated to some extent by using automatic tissue segmentation (e.g., the white and gray brain matter) using openly available software[92]. Neural network approaches for CEST reconstruction can also incorporate uncertainty quantification[93], which provides additional information on the reliability of the predicted output.

Finally, the last and most challenging line of validation is human patient population, where the limited sources for ground truth are the final clinical outcome, biopsies (if available), and conventional imaging protocols (such as Gd-enhanced, DWI, perfusion, MRS, etc.).

Advantages and limitations

In summary, deep saturation transfer MRF offers a rapid means to acquire and reconstruct quantitative molecular information in-vivo. Moreover, as existing CEST-MRF methods do not yet make full usage of k-space under-sampling (as in conventional MRF[33,94]) it has the potential of even higher accelerations in the future. Compared to CEST-weighted imaging, it has the advantage of disentangling the contributions of various compounds and contrast mechanisms. It provides faster acquisition and higher spatial resolution compared to MRS and, in contrast to PET, does not use ionizing radiation. The main limitation of CEST-MRF is the lack of *in-vivo* validation references, where it is challenging (or sometimes impossible) to obtain ground truth estimates of the exact physical properties. As a result, the validation of CEST-MRF for new applications or diseases may require the acquisition of additional reference information ("best standard available", e.g., using 31P MRS-based pH measurements, QUESP measurements of proton exchange rate, and tissue biopsy assays), which are lengthy and cumbersome. Another limitation lies in the lengthy dictionary generation step, which can take several hours or more for multi-pool, pulsed-wave acquisition scenarios[29]. While neural networks can be applied for further acceleration of the dictionary generation step[43,95] they still need to be extensively pre-trained for new molecular targets.

Reagents

Equipment

Materials and Equipment

- Access to a preclinical or a clinical MRI scanner and certified staff is required. A static magnetic field of 3T and higher is recommended for CEST MRF applications.

- For preclinical rodent imaging the following items are needed:

- A vaporizer

- Isoflurane

- Nose cone

- Rectal temperature monitor

- Hot air blower or an animal cradle with built in hot water circulation

- Animal respiration monitor

Computational hardware

A desktop/laptop computer with CUDA-enabled GPU is required. The minimum GPU required is Nvidia RTX 1050 (or equivalent). All the experiments described in this paper were tested using a desktop computer with the following hardware: Intel I9-12900F processor, Nvidia RTX 3060 12 GB GPU, 32 GB RAM. Alternatively, the computational pipeline can run on the cloud using Google Colab (https://colab.google/), which provides a free virtual machine with a GPU service that can load the Jupyter notebooks provided.

Computational software

The code can run on any operational system although Linux is recommended. Python version 3.9 needs to be installed. We recommend using Conda/Miniconda for python environment management (https://docs.anaconda.com/free/anaconda/install)or venv (for advanced users). The molecular MRF package can also be installed as a docker image (https://hub.docker.com/repository/docker/vnikale/pycest-starter/general) following docker installation (https://docs.docker.com/engine/install/).

(Optional) For *in-vitro* imaging

in preclinical/clinical scanners, the following items are required:

- pH meter

- Vortex mixer

- Weighing balance (milligram range sensitivity)

- 50 mL Falcon test tubes (preclinical) / 1.8L water cooler (Coleman, USA) (clinical)

- Saline (2L)

- 2 mL glass screw top vials (preclinical) / 15mL test tubes (clinical)

- PBS

- A vial holder for clinical scanner imaging. A 3D printer CAD (.stl) file is provided in the folder supplementary/phantom_cad in https://github.com/momentum-laboratory/molecular-mrf.

- Compound(s) of interest (e.g. L-arginine, CAS 74793, Merck)

- Pipettes

- NaOH and HCl for pH titration

Procedure

(Optional) *In-vitro* phantom preparation ● Timing 3h

As discussed earlier, in-vitro phantoms can be assembled as a preliminary means to assess the parameter quantification performance. A variety of CEST-phantoms can be used for this purpose, depending on the target application (e.g., bovine serum albumin (BSA) for brain imaging, creatine and phosphocreatine for muscle metabolism imaging, etc). While instructions for CEST-MRI phantom preparation can be found elsewhere (e.g., at[57,58,96]), we provide here a simple-to-make L-arginine phantom example, which will enable reproducing the results shown in **Fig. 6.**

!CAUTION Proper safety techniques should be followed when handling chemicals. All phantom preparations should be carried out with caution in a fume hood. Use protective eye wear, gloves and a lab coat.

1. Based on the molecular weight of the compound of interest (174.20 gr/mol for L-arginine), and the volume of the intermediate preparation container (e.g., a 50 mL Falcon tube), use the spatula, weighing paper, and balance to weigh the required powder needed for the largest sample concentration (e.g., 100 mM).

2. Add PBS 1x and thoroughly stir the closed container using the vortex mixer. Visually verify the solubility of the sample.

3. Titrate the sample using the NaOH/HCl and the pH meter into the desired pH levels (for the L-arginine example shown in **Fig. 6**, use pH 4, 4.5, 5, 5.5, or 6), while stirring between measurements.

4. Pour the resulting solution into a 2 mL glass vias (for preclinical phantoms) or 15 mL test tube (for clinical-scanner-oriented phantoms).

5. Use the remaining solution in the 50 mL falcon tube to synthesize additional samples by serial dilution. Titrate to reach the desired pH.

6. After completing three 2mL glass vial samples (preclinical) or six 15 mL test tube samples (clinical), position the samples within a new 50 mL Falcon tube (preclinical) or a dedicated test tube holder (clinical). See setup examples in **Fig. 6** (right column).

7. Fill the large phantom container completely using PBS 1x or saline. Minimize the existence of any air bubbles.

Pulse sequence setup

8. CEST MRI simulator installation: ● Timing 5-10 min

Download the signal simulator from the repository https://github.com/momentum-laboratory/molecular-mrf and extract it. If you have git installed, you can use the following command instead:

```
git clone git@github.com:momentum-laboratory/molecular-mrf.git
```

The prerequisites are C/C++11 compiler, SWIG and Python version v3.9 or newer. While C/C++11 compiler is installed by default in most Linux distributions, it requires installation on Windows and Mac (e.g., using https://www.cygwin.com/ or https://visualstudio.microsoft.com/). On Linux and Mac OS machine SWIG can be installed using a default package manager, for example:

```
sudo apt-get install swig
```

On a Windows machine it can be downloaded from https://www.swig.org/download.html. For the next steps of the installation, it is important to create a clean environment for python v3.9, using conda or venv:

```
conda create –n name_of_env python=3.9

conda activate name_of_env
```

Change the current working directory to the repository folder and install the simulator:

```
cd open-py-cest-mrf

pip install –e .
```

The flag –e allows installation in editable mode and the (.) is necessary. ▲CRITICAL The installation folder path is fixed. If the package is moved re-installation will be required. To verify the success of the installation, run:

```
pip list
```

You should see a variety of packages, including the following critical two:

```
BMCSimulator 0.2

cest-mrf 0.2 .../open-py-cest-mrf
```

TROUBLESHOOTING.

If the installation failed, you can employ docker and run the following command:

```
docker pull vnikale/pycest-starter
```

9. Loading and installing CEST-MRF pulse sequences into the MRI scanner ● Timing 5 min

For preclinical (Bruker MRI) scanners, the complete pulse sequence should be downloaded from https://osf.io/52bsg and installed according to the detailed manual provided within the link. For clinical scanners utilizing the pulseq framework, the .seq files should be loaded into the scanner computer, as detailed in the pulseq installation manual (http://pulseq.github.io/, typically into the folder c:\MedCom\Mricustomer\seq\pulseq\). The compound specific pulse sequences (**Table 1**) are provided in the folder 'supplementary/published_pulse_sequences/' in https://github.com/momentum-laboratory/molecular-mrf and in the pulseq CEST open library https://github.com/kherz/pulseq-cest-library/tree/master/seq-library.

10. Creating new CEST-MRF pulse sequences (example given for L-arginine) ● Timing 30 sec

The previous step enables the use of previously published/generated MRF pulse sequences. For research purpose, one may wish to adjust and optimize CEST-MRF for a new compound target, or a different imaging scenario. As an example, we will define an MRF sequence for L-arginine quantification[30,73] using a continuous wave saturation pulse at 9.4T. The Code snippets provided below can also be found in the “dot_product_example” folder (within the main https://github.com/momentum-laboratory/molecular-mrf repository).

First, the pypulseq package needs to be imported:

```
import pypulseq
```

Next, the acquisition parameters are defined. In this example, the sequence consists of a saturation pulse fixed at the L-arg chemical shift (3.0 ppm), with 30 images acquired with varying saturation pulse powers (B_1):

```
seq_defs = {}

seq_defs['n_pulses'] = 1 # number of pulses

seq_defs['tp'] = 3 # pulse duration [s]

seq_defs['td'] = 0 # inter-pulse delay [s]

seq_defs['Trec'] = 1 # delay before readout [s]

seq_defs['DCsat'] = seq_defs['tp'] / (seq_defs['tp'] + seq_defs['td']) # duty cycle

seq_defs['offsets_ppm'] = [3.0] * 30 # offset vector [ppm]


seq_defs['num_meas'] = len(seq_defs['offsets_ppm']) # number of images

seq_defs['Tsat'] = seq_defs['n_pulses'] * (seq_defs['tp'] + seq_defs['td']) - seq_defs['td']

seq_defs['B0'] = 9.4 # B0 [T]

seq_defs['seq_id_string'] = 'larg_seq’ # unique seq id

seq_defs['B1pa'] = [5, 5, 3, 3.75, 2.5, 1.75, 5.5, 6, 3.75,5.75, 0.25, 3, 6, 4.5, 3.75, 3.5, 3.5, 0, 3.75, 6, 3.75, 4.75, 4.5, 4.25, 3.25, 5.25, 5.25, 0.25, 4.5, 5.25]
```

For the readout, we will use a 60 degrees flip angle followed by signal ADC:

```
imaging_pulse = pypulseq.make_block_pulse(60 * np.pi / 180, duration=2.1e-3)

imaging_delay = pypulseq.make_delay(20e-3)

pseudo_adc = pypulseq.make_adc(1, duration=1e-3)

seq = pypulseq.Sequence()
```

11. Now we can build the entire sequence in a loop which calls the various definitions declared earlier:

```
for idx, B1 in enumerate(seq_defs['B1pa']):

if idx > 0: # add relaxtion block after first measurement

seq.add_block(pypulseq.make_delay(seq_defs['Trec'] - te)) # net recovery time

# saturation pulse

current_offset_hz = seq_defs['offsets_ppm'][idx] * seq_defs['B0'] * gyro_ratio_hz

fa_sat = B1 * gyro_ratio_rad * seq_defs['tp'] # flip angle of sat pulse

# add pulses

for n_p in range(seq_defs['n_pulses']):

# If B1 is 0 simulate delay instead of a saturation pulse

if B1 == 0:

seq.add_block(pypulseq.make_delay(seq_defs['tp'])) # net recovery time

else:

sat_pulse = pypulseq.make_block_pulse(fa_sat, duration=seq_defs['tp'], freq_offset=current_offset_hz)

seq.add_block(sat_pulse)

# delay between pulses
```

```
if n_p < seq_defs['n_pulses'] - 1:

seq.add_block(pypulseq.make_delay(seq_defs['td']))


# Imaging pulse

seq.add_block(imaging_pulse)

seq.add_block(imaging_delay)

seq.add_block(pseudo_adc)
```

The definitions are now stored and added as human-readable text to the final pulse sequence (.seq) file:

```
def_fields = seq_defs.keys()

for field in def_fields:

seq.set_definition(field, seq_defs[field])

seq.write(seq_fn)
```

▲CRITICAL Note that the output .seq file serves a dual purpose, providing instructions for *both* the in-silico simulator and the physical scanner. For hybrid pulseq/vendor protocols, where the readout is separately determined at the scanner, separated files should be created (see an example in the function 'write_sequence_clinical', located in the folder 'dot_product_example' from the provided GitHub repository).

▲CRITICAL If you prefer using .py scripts instead of Jupyter notebooks make sure to run them from the root folder as modules. Otherwise, key utility functions will not be accessible. For example, to run the dot_prod_example, use:

```
python -m dot_prod_example.preclinical
```

Animal / human subject setup ● Timing 10 min / 5 min per animal / human subject, respectively

This part of the protocol has two main variants: (i) Animal (rodent) setup in preclinical scanners or (ii) Human setup in clinical scanners. In both cases, the steps are not unique for CEST-MRF, and follow traditional MRI research practice[97,98,99].

Animal positioning

▲CRITICAL All procedures involving animal use must be performed in accordance with institutional and national guidelines and regulations and approved by the relevant animal care and use committees.

12. Place the rodent in an induction chamber containing 4% wt/vol isoflurane with air or oxygen. Upon relaxation, transfer the animal to the dedicated scan cradle, while gradually reducing the isoflurane level to ~1–2% wt/vol , to be applied via a nose cone.

! CAUTION Isoflurane must be scavenged properly (e.g., using a charcoal canister or vacuum pumps) to minimize exposure to humans.

13. Gently fix the organ to be imaged using dedicated bars, tape, or small cushions.

14. Apply lubricant eye ointment to prevent drying of the cornea.

▲CRITICAL An MRI-compatible respiration rate monitoring sensor should be used to ensure mouse well-being.

15. Maintain the animal body temperature at 37°C using a hot air blower, circulating water pump, or similar, while using a rectal temperature sensor for feedback.

16. Position the animal cradle within the RF coil and make sure the target organ is at the center of the magnet.

Human subject setup

! CAUTION All procedures involving human imaging must be performed under approval by the local ethics/IRB committee and in accordance with institutional and national guidelines and regulations. Written informed consent must be given before the study.

17. Gently position the subject on the scanner bed and install the organ-relevant coil.

18. Verify the use of ear-plugs, and the subject familiarity with the hand-pump used to communicate with the staff in case of discomfort or emergency.

19. Position the isocenter at the middle of the organ of interest using the laser marking.

! CAUTION Make sure to notify the subject to close their eyes prior to activating the laser.

20. Slowly insert the subject into the magnet.

Raw data acquisition ● Timing 12 - 30 min per subject/animal (depending on optional steps)

21. Acquire scout images and calibrate the RF coils (typically performed manually in preclinical scanners using the matching and tuning rods).

22. Determine the location of the slice of interest (axial orientation is recommended to minimize B_0 inhomogeneity) or slice stack. A minimal voxel size of ~0.3 mm x 0.3 mm x 1 mm is recommended for rodents, and 1.8 mm size isotropic for humans.

23. Shim the magnetic field manually or using vendor-provided automatic shimming.

24. Scan the subject/animal using the CEST-MRF pulse sequence of interest (see **Table 1** for a detailed list of available pulse sequences).

25. **(Optional)** Acquire T_1, T_2, B_0, and/or B_1 maps (using vendor-provided protocols) while using the same image geometry used in the CEST-MRF pulse sequence. These maps are required for implementing the sequential learning approach (**Fig. 4**

and **Fig. 7**), or as a means for later exploring the origins of noisy MRF reconstruction (such as those stemming from inhomogeneous B_0 or B_1 fields). For B_0/B_1 mapping we recommend the use of the water saturation shift referencing (WASSR[100]) sequence or the simultaneous mapping of water shift and B_1 (WASABI[101]) sequence, as provided in the pulseq-CEST open library (https://github.com/kherz/pulseq-cest-library/tree/master/seq-library).

26. **(Optional)** Acquire additional contrast-weighted images for comparison. These may include T_2-weighted images, diffusion, and perfusion images, depending on the pathology of interest and using vendor-provided pulse sequences. Traditional CEST Z-spectrum can be acquired by using the pulse sequence files: APT_3T_00x.seq, available at https://github.com/kherz/pulseq-cest-library/tree/master/seq-library, where x is replaced by 0-4 in order of the sequence effectiveness, as described in the CEST-weighted brain imaging consensus paper[102]. A preclinical CEST-weighted pulse sequence for Bruker scanners is given in the 'supplementary/published_pulse_sequences/cest_weighted' folder at https://github.com/momentum-laboratory/molecular-mrf/.

Image analysis, processing, and AI-based reconstruction

27. CEST MRF dictionary generation ● Timing 10 sec – 22 hrs (depends on the number of CPU available and the scenario of interest)

To generate synthetic signal trajectories (exemplified here for L-arginine quantification at 9.4T) a .yaml file which stores the required magnetic tissue properties is required. These definitions are determined by the user in the 'configs.py' file:

```
class ConfigPreclinical(Config):

def __init__(self):

config = {}

config['yaml_fn'] = 'scenario.yaml'

config['seq_fn'] = 'acq_protocol.seq'

config['dict_fn'] = 'dict.mat'


# Water_pool

config['water_pool'] = {}

config['water_pool']['t1'] = np.arange(2500, 3350, 50) / 1000

config['water_pool']['t1'] = config['water_pool']['t1'].tolist() # vary t1

config['water_pool']['t2'] = np.arange(600, 1250, 50) / 1000

config['water_pool']['t2'] = config['water_pool']['t2'].tolist() # vary t2

config['water_pool']['f'] = 1
```

```
# Solute pool

config['cest_pool'] = {}

config['cest_pool']['Amine'] = {}

config['cest_pool']['Amine']['t1'] = [2800 / 1000]

config['cest_pool']['Amine']['t2'] = [40 / 1000]

config['cest_pool']['Amine']['k'] = np.arange(100, 1410, 10).tolist()

config['cest_pool']['Amine']['dw'] = 3

config['cest_pool']['Amine']['f'] = np.arange(10, 125, 5) * 3 / 110000

config['cest_pool']['Amine']['f'] = config['cest_pool']['Amine']['f'].tolist()

...

config['b0'] = 9.4

config['num_workers'] = 16
```

Note that python list definitions (.tolist()) are needed for flawless .yaml saving. The variable num_workes determines how many threads will be used (depending on the number of available CPUs). Finally, the .yaml file is generated and saved using the function:

```
write_yaml_dict(cfg, yaml_fn)
```

Based on the .yaml tissue parameter combinations and the desired acquisition protocol defined in the .seq file, the dictionary simulation can take place using the following function:

```
dictionary =
generate_mrf_cest_dictionary(seq_fn=seq_fn,param_fn=yaml_fn,dict_fn=dict_fn,num_workers=cfg.num_workers,axes='xy')
```

The input arguments *_fn expect strings with the names of the .yaml, .seq, and .mat files, respectively. The output dictionary that stores all synthetic signal trajectories and their paired parameters will be saved as a .mat file. The parameter ‘axes’ selects which magnetization vector component is stored. While the default is ‘xy’, the ‘z’ option is useful for Z-spectrum simulation where no readout is simulated. For credible simulation, the practical limitations of the scanner should be explicitly declared, for example:

```
...
lims = pp.Opts(
max_grad=40,
grad_unit="mT/m",
max_slew=130,
slew_unit="T/m/s",
rf_ringdown_time=30e-6,
rf_dead_time=100e-6,
rf_raster_time=1e-6,
gamma=cfg.gamma/2/np.pi*1e6,
)
...
gx_spoil, gy_spoil, gz_spoil = [
pp.make_trapezoid(channel=c, system=lims, amplitude=spoil_amp, duration=spoil_dur, rise_time=rise_time)
...
```

28. Dot-product matching of L-arginine phantom data ● Timing 10 sec (L-arginine) – 2.3 hrs (in-vivo)[29]

As we now possess both acquired data and a simulated dictionary, parameter quantification can take place. The function for dot product matching is available at the repository folder 'open-py-cest-mrf/cest_mrf/metrics/dot_product.py':

```
def dot_prod_matching(dictionary = None, acquired_data = None, dict_fn = None, acquired_data_fn = None, batch_size = 256):
...
```

The function can either receive the dictionary and the acquired data as variables or strings containing the respectively saved file names ('dict_fn', 'acquired_data_fn'). Dot product matching is performed in batches due to RAM considerations and can be adjusted according to the available hardware using the argument 'batch_size'. ▲CRITICAL The number of image pixels must be divisible by 'batch_size'.

The next steps include image reshaping, L_2 normalization, and dot-product calculation in batches:

```
data = acquired_data.reshape((n_iter, r_raw_data * c_raw_data), order='F')


norm_dict = synt_sig / (la.norm(synt_sig, axis=0) + 1e-10)

norm_data = data / (la.norm(data, axis=0) + 1e-10)


batch_indices = range(0, data.shape[1], batch_size)

for ind in range(np.size(batch_indices)):

current_score = norm_data[:, batch_indices[ind]: batch_indices[ind] + batch_size].T @ norm_dict

dp[0, batch_indices[ind]: batch_indices[ind] + batch_size] = np.max(current_score, axis=1)

dp_ind = np.argmax(current_score, axis=1)


t1w[0, batch_indices[ind]: batch_indices[ind] + batch_size] = dict_t1w[0, dp_ind]

…
```

For each batch, the maximal dot-product is found, and its corresponding index is stored. An example dot-product matching for the provided L-arginine phantom data is performed using the following command:

```
quant_maps =

dot_prod_matching(dict_fn='dict.mat', acquired_data_fn='acquired_data.mat')
```

The resulting quantitative exchange parameter maps are shown in **Fig. 6**.

29. Supervised deep reconstruction network training (exemplified for L-arg data) ● Timing 5 min (L-arg) – 6 hrs (in-vivo)[29]

The full code for this step can be found in the jupyter notebook file deep_reco_example/deep_reco_preclinical.ipynb. First PyTorch is installed:

```
!pip3 install torch torchvision –index-url https://download.pytorch.org/whl/cu118
```

TROUBLESHOOTING

▲CRITICAL: If the user prefers using the .py script instead of the Jupyter notebook, PyTorch and OpenCV. Need to be installed using:

```
pip3 install torch torchvision --index-url https://download.pytorch.org/whl/cu118
```

```
pip3 install opencv-python
```

After importing all required libraries, a fully connected NN is defined:

```
class Network(nn.Module):

def __init__(self):

super(Network, self, sig_n).__init__()

self.l1 = nn.Linear(sig_n, 300)

self.relu1 = nn.ReLU()

self.l2 = nn.Linear(300, 300)

self.relu2 = nn.ReLU()

self.l3 = nn.Linear(300, 2)
```

```
def forward(self, x):

x = self.l1(x)

x = self.relu1(x)

x = self.l2(x)

x = self.relu2(x)

x = self.l3(x)

return x
```

The network receives an input signal of length sig_n and returns a vector of two parameters: concentration and proton exchange rate. The Pytorch Dataset class is used to import signals throughout the training from the dictionary .mat file created earlier.

▲CRITICAL For simplicity, validation datasets are not described here, yet they are required for robust learning. A simple first line of validation can be performed using separately simulated signal trajectories with parameter combinations that were not present in the training dictionary.

In the next step we define the training hyper-parameters:

```
learning_rate = 0.0001

batch_size = 256

num_epochs = 100

noise_std = 0.002 # noise level for training
```

As NN converge faster and more efficiently with normalized data, we will next find the minimum and maximum values of the output exchange parameters (as set in the dictionary):

```
temp_data = sio.loadmat('dict.mat')

min_fs = np.min(temp_data['fs_0'])

min_ksw = np.min(temp_data['ksw_0'].transpose().astype(np.float))

max_fs = np.max(temp_data['fs_0'])
```

```
max_ksw = np.max(temp_data['ksw_0'].transpose().astype(np.float))


min_param_tensor = torch.tensor(np.hstack((min_fs, min_ksw)), requires_grad=False)
max_param_tensor = torch.tensor(np.hstack((max_fs, max_ksw)), requires_grad=False)
```

Now let’s initialize the network, optimizer and data loader:

```
reco_net = Network(sig_n).to(device)


optimizer = torch.optim.Adam(reco_net.parameters(), lr=learning_rate)


training_data = sio.loadmat('dict.mat')
dataset = DatasetMRF(training_data)


train_loader = DataLoader(dataset=dataset,
batch_size=batch_size,
shuffle=True,
num_workers=8)
```

After the target normalization the training loop takes place using the mean-squared-error loss:

```
...
target = torch.stack((cur_fs, cur_ksw), dim=1)


target = normalize_range(original_array=target, original_min=min_param_tensor,
original_max=max_param_tensor, new_min=-1, new_max=1).to(device).float()
...
loss = torch.mean((prediction - target) ** 2)
```

30. Supervised reconstruction network inference (exemplified using L-arg data) ● Timing 10 sec

The training will stop after reaching the maximum number of epochs set and the model will be stored as a checkpoint. A previously stored checkpoint can be loaded using the following lines:

```
loaded_checkpoint = torch.load('checkpoint')

reco_net = Network(sig_n).to(device)

reco_net.load_state_dict(loaded_checkpoint['model_state_dict'])
```

Next, the experimentally acquired data is loaded, and inference takes place:

```
acquired_data = sio.loadmat('acquired_data.mat')['acquired_data'].astype(np.float)

_, c_acq_data, w_acq_data = np.shape(acquired_data)


# Reshaping the acquired data to the shape expected by the NN (e.g. 30 x ... )

acquired_data = np.reshape(acquired_data, (sig_n, c_acq_data * w_acq_data), order='F')

acquired_data = acquired_data / np.sqrt(np.sum(acquired_data ** 2, axis=0))


# Transposing for compatibility with the NN - now each row is a trajectory

acquired_data = acquired_data.T


acquired_data = torch.from_numpy(acquired_data).to(device).float()

acquired_data.requires_grad = False


# Switching to evaluation mode

reco_net.eval()


prediction = reco_net(acquired_data)


prediction = un_normalize_range(prediction, original_min=min_param_tensor.to(device),
```

```
original_max=max_param_tensor.to(device), new_min=-1, new_max=1)
```

Note that this step has also to un-normalize the output parameters back into physically meaningful units.

31. Sequential reconstruction network training and inference (exemplified using Iohexol phantoms and tumor bearing mice) ● Timing 10-15 min

The sequential reconstruction pipeline is very similar to the previously described reconstruction network. The primary difference lies in the network ability to receive as input the raw MRF signals (either simulated or experimental) alongside previously determined parameters, such as water T_1 and T_2 maps:

```
...

target = torch.stack((cur_fs, cur_ksw), dim=1)

target = normalize_range(original_array=target, original_min=min_param_tensor,

original_max=max_param_tensor, new_min=0, new_max=1).to(device)


input_water_t1t2 = torch.stack((cur_t1w, cur_t2w), dim=1)

input_water_t1t2 = normalize_range(original_array=input_water_t1t2, original_min=min_water_t1t2_tensor,
original_max=max_water_t1t2_tensor, new_min=0, new_max=1)

...

noised_sig = torch.hstack((input_water_t1t2, noised_sig))

prediction = reco_net(noised_sig.to(device).float())
```

The folder "sequential_example" includes two step-by-step examples of training sequential networks, with the expected output maps shown in **Fig. 7**. Note that the supplementary input of T_1 and T_2 maps can be augmented by B_0 and B_1 field maps, or any other separately determined properties. The process can continue recursively using the exact same approach; e.g., by leveraging the semisolid MT exchange parameters output from a first network as input for the next[29].

TROUBLESHOOTING

32. Multi-pool deep reconstruction of human data● Timing 10 sec

The pulse sequences used in previously published human molecular MRF studies are provided here as open .seq files (**Table 1**). A fully trained neural network for reconstructing raw CEST-MRF data, acquired on a 3T scanner with an EPI readout, into six quantitative parameter maps is available under the human_example folder at https://github.com/momentum-laboratory/deep-molecular-mrf alongside a raw MRF human data and a full inference code. The training pipeline for this clinical network is very similar to the supervised deep reconstruction examples given above. For specific training parameters see[41,67]. The expected output for the quantitative parameter map reconstruction is shown in **Fig. 8**.

33. (Optional alternative) Unsupervised training and reconstruction of semisolid MT and water parameter maps ● Timing 10 sec for inference, ~22 hours for training[40]

In case the user has access to several healthy human volunteers or patients and would rather train the reconstruction network on real acquired data instead of simulated dictionaries, the unsupervised reconstruction route may be chosen, replacing the need to generate dictionaries. Note that the acquisition is still based on MRF principles (**Fig. 5**). The pulse sequence used in a previously published unsupervised human semisolid MT MRF study[40] is provided here (**Table 1**). A fully trained neural network for reconstructing semisolid MT MRF data, acquired on a 3T scanner into four quantitative parameter maps is available under the unsupervised_example folder at https://github.com/momentum-laboratory/deep-molecular-mrf, alongside a raw MRF human data and a full inference code. The function can be activated using the following command:

```
python -m unsupervised_example.cnn_inference
```

where ‘data’ and ‘result’ are strings containing the paths to the acquired data and the folder where the results will be saved, respectively. ‘model’ is a string containing the path for the trained model. For specific training parameters see[40]. The expected output for the unsupervised parameter map reconstruction is shown in **Fig. 9**.

Data availability

All the data used in this work is available at https://github.com/momentum-laboratory/deep-molecular-mrf. It includes raw MRF data, quantitative parameter maps (**Figs. 6-9**), a CAD file for 3D printing a six vial (phantom) holder, and pulse sequence files (**Table 1**). A complete preclinical CEST-MRF pulse sequence for Bruker scanners is available at https://osf.io/52bsg. The .seq format files used in this work were also deposited at the pulseq CEST open library: https://github.com/kherz/pulseq-cest-library/tree/master/seq-library.

Code availability

All code is available on https://github.com/momentum-laboratory/deep-molecular-mrfin the format of Python scripts and Jupyter notebooks.

Author contributions statement: Conceptualization: N.V., C.T.F, and O. P. Methodology: N.V., O.C., H.Y.H, M.Z., C.T.F, and O.P. Data curation: O.C., H.Y.H., M.Z., C.T.F, and O.P. Writing: N.V. H.Y.H, C.T.F, and O.P. Reviewing and editing: N.V., O.C., H.Y.H., M.Z., C.T.F., and O.P. Supervision: C.T.F. and O.P.

Competing interests: The authors declare the following competing interests: C.T.F. and O.C. hold a patent for the CEST MR fingerprinting method (patent no. US10,605,877).

Troubleshooting

See supplementary table 2

Time Taken

The entire protocol takes ~48 min - 57 hrs to complete depending on the optional steps and the imaging scenario of interest.

1. In-vitro phantom preparation (**optional**) - 3h
2. Pulse sequence setup -10.5 – 15.5 min

- CEST MRI simulator installation - 5-10 min
- Loading and installing the pulse sequences into the MRI scanners 5 min
- CEST MRF sequence file creation - 30 sec

1. Animal / human subject setup - 10 min / 5 min: per animal / human subject
2. Raw data acquisition – 35 sec – 30 min (**depends on the optional steps**)
3. Image analysis, processing, and AI-based reconstruction - ~16 min – 52.6 hrs

- CEST MRF dictionary simulation - 10 sec – 22hrs *
- Dot-product matching - 10 sec – 2.3 hrs*
- Supervised reconstruction network training - 5 min – 6 hrs*
- Supervised reconstruction network inference (including variable preparation and loading) - 10 sec.
- Sequential reconstruction network training and inference - 10-15 min
- Multi-pool inference of human parameter maps (including variable preparation and loading) - 10 sec.
- Self-supervised learning pipeline (optional alternative) – 10 sec -22 hrs

*The timing is given for L-arginine phantom data (CW pulses with ~665K dictionary entries). Changing the imaging scenario of interest may result in drastic timing changes, depending on the dictionary size, saturation pulse properties (CW/PW) and the number of simulated pools. For example, generating a dictionary of 273,790 entries, acquired using a PW clinical sequence (see the dp_clinical case folder in the provided GitHub repository) took 190s. A previous work reported the generation of ~70M entries for a multi-pool pulsed sequence in ~22 hours and the corresponding dot-product matching (for a single slice image) taking ~2.3 hours[29].

** The inference takes about 10 s. A previous work reported that training using 5 subject data and 153 slices took ~22 hours[40].

Anticipated Results

The protocol outlines all steps needed for conducting semisolid MT/CEST MRF experiments, yielding seven main output file-types (italicized in the following paragraphs). Initially, in-vitro phantoms are assembled (**Fig. 6,** right column). Next, *pulse sequence instruction files* are generated (.seq and .txt formats) as well as a *descriptor file* for the biological scenario of interest and its associated multi proton pool parameters (.yaml format). The pulse sequence instruction files are then loaded into the MRI scanner, where *raw MRF data* are acquired from phantoms, animals, or humans (.dicom or .npy format). Using the .seq and .yaml files, a simulated imaging experiment takes place in-silico, generating a *dictionary* of synthetic signal trajectories and their paired magnetic properties (.mat format).

Quantitative image mapping can then be performed using the experimentally acquired raw data and the simulated signal dictionaries based on dot-product matching. The output will be a set of *reconstructed images* (.mat and .eps formats). Deep learning-based reconstruction is typically faster and recommended, generating an *optimized set of neural network node weights* (checkpoint.pt format), and *reconstructed parameter maps* (.mat and .eps formats).

The provided GitHub repository includes five main exemplary case studies (L-arginine, Iohexol, tumor-bearing mouse, supervised, and unsupervised human imaging), arranged into separate folders. Each folder contains sample raw data, a Jupyther notebook (.ipynb), and a python script (.py) file. The provided scripts enable the generation of the above-mentioned results and reproducing the images shown in **Fig. 6**, **Fig. 7**, **Fig. 8**, and **Fig. 9**.

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Figures

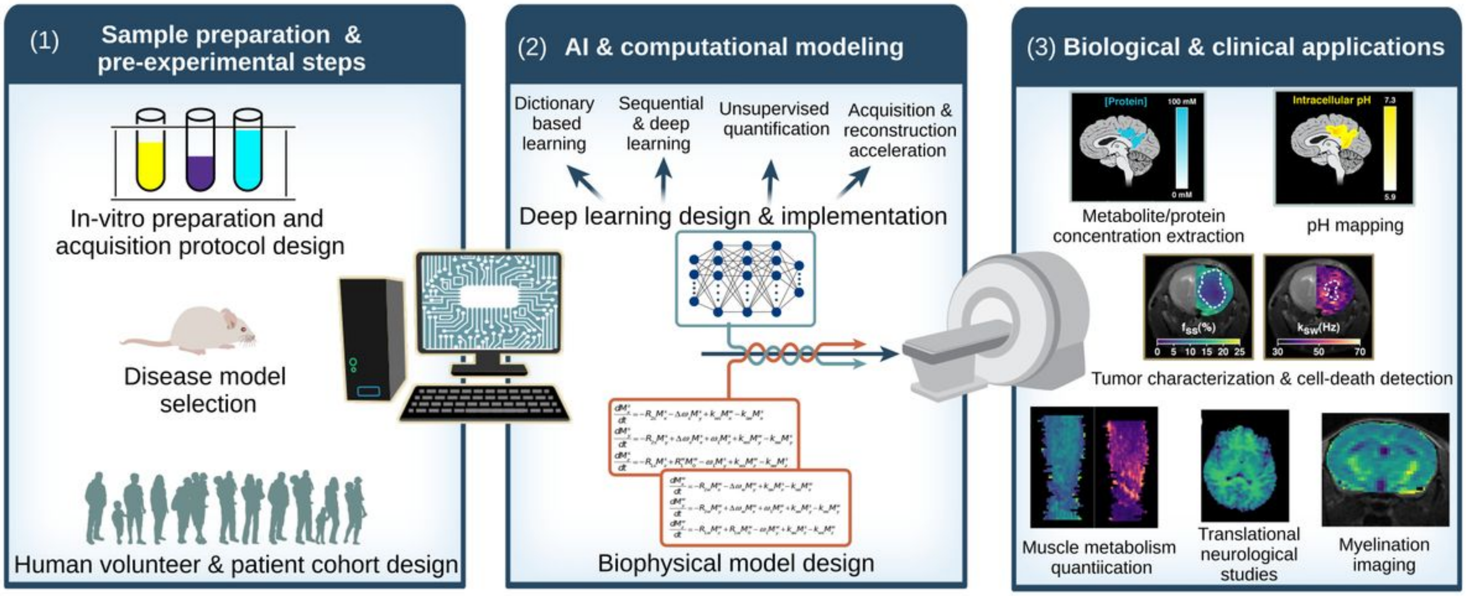

Figure 1

Fig. 1. Practical overview of the deep molecular magnetic resonance fingerprinting protocol. The workflow consists of three main stages: (1) Sample/subject preparation and acquisition protocol design. (2) Computational modeling and AI training, and (3) Evaluation and interpretation for the biological/clinical scenario of interest. Sub-figures in step (3) are reproduced with permission from Perlman et al. *Nat. Biomed. Eng*. 2022[29], Weigand-Whittier et al., *Magn. Res. Med*. 2023[42], and Perlman et al., *Magn. Res. Med*. 2022[47].

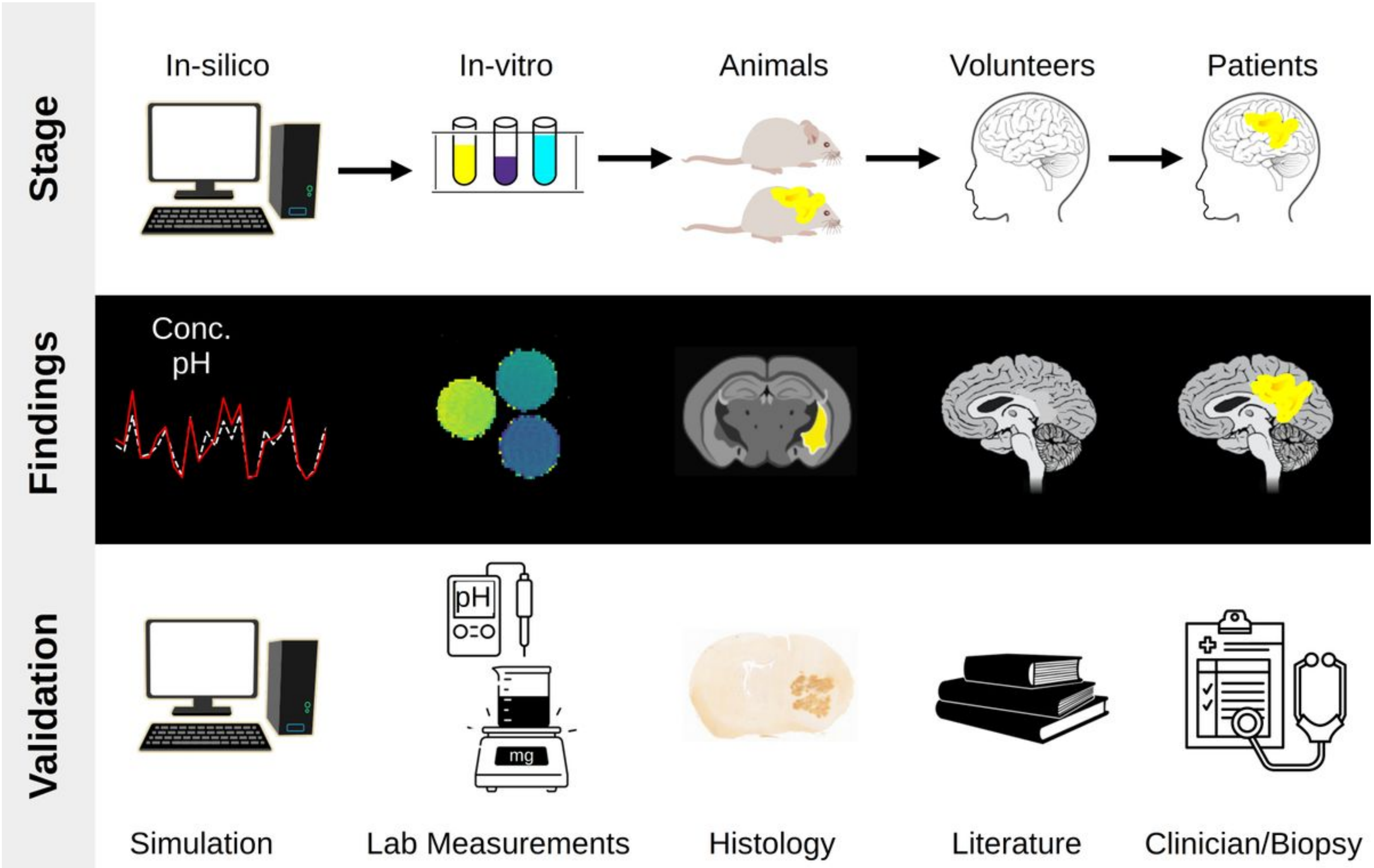

Figure 2

Fig. 2. Experimental validation pipeline for CEST-MRF. Each column represents different biological settings, in an ascending order of complexity (left to right). (**Top**). Experimental stage. (**Center**). Expected CEST-MRF findings. (**Bottom**). Validation means. The histology sub-figure is reproduced with permission from Perlman et al. *Nat. Biomed. Eng*. 2022[29].

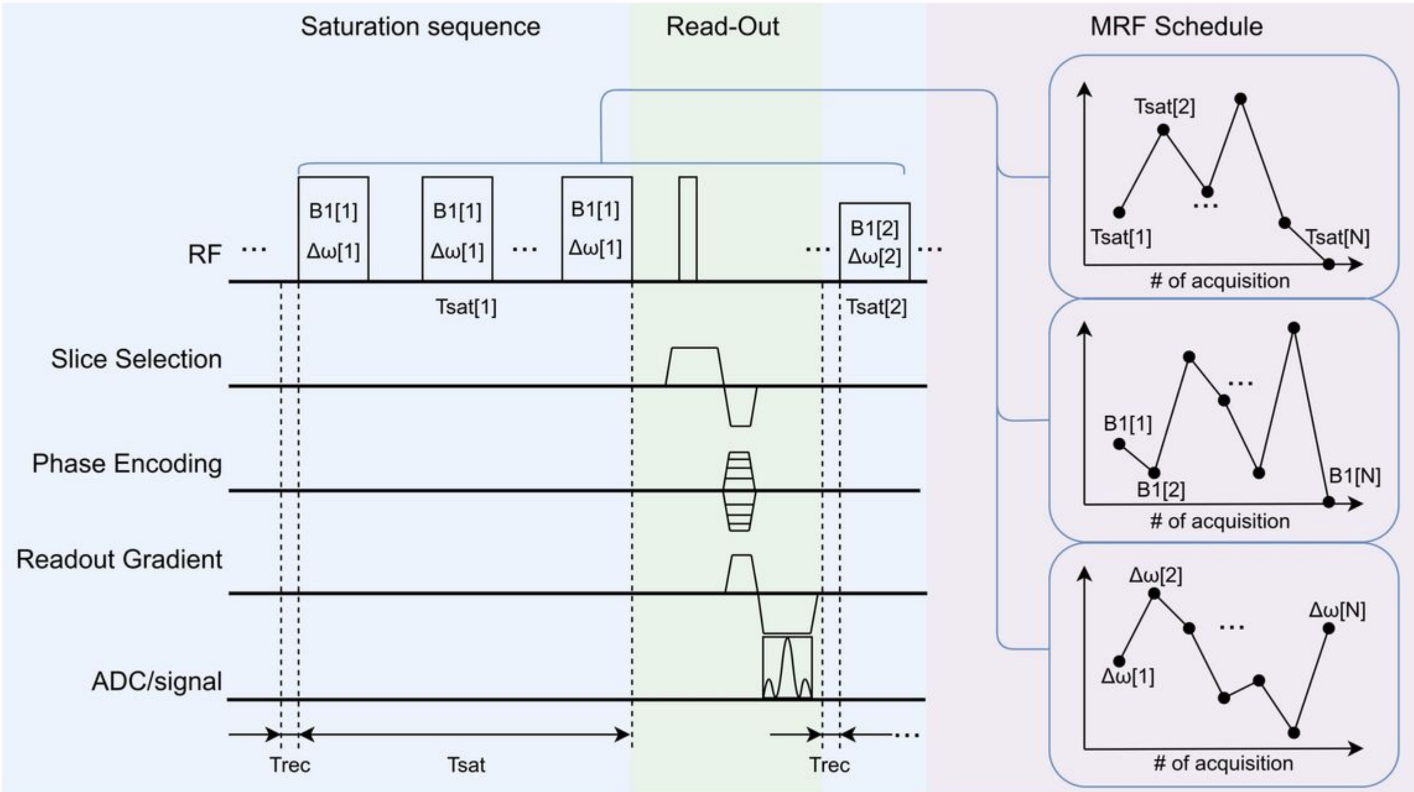

Figure 3

Fig. 3. A typical CEST MRF pulse sequence diagram. It consists of a frequency-selective saturation pulse train, a rapid read-out module (e.g., 3D EPI, turbo spin echo, etc.), recovery and delays. The sequence is repeated N times with varied saturation pulse parameters, yielding N raw MRF images.

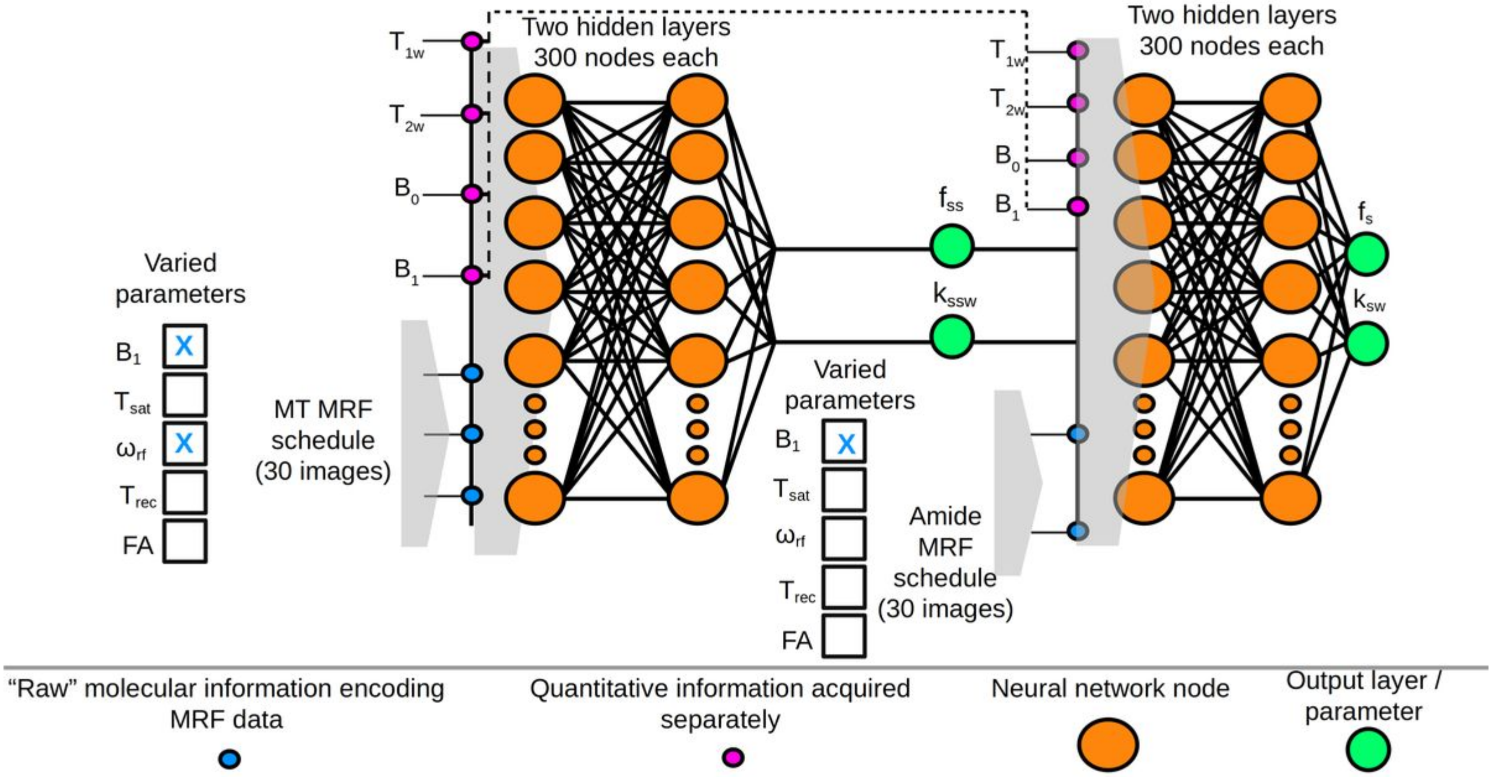

Figure 4

Fig. 4. Sequential learning of multi-proton-pool information. A semisolid MT-oriented MRF acquisition schedule yields raw images that are fed voxel-wise into the first NN, together with the quantitative water pool and field homogeneity maps (T_1, T_2, B_0, B_1). This NN maps the semisolid MT pool exchange parameters, which are then fed, together with the previously obtained quantitative data, into the second NN, ultimately yielding the proton exchange parameters (f_s and k_{sw}) of the compound of interest. Reproduced and modified with permission from Perlman et al*., NMR Biomed*. 2023[84].

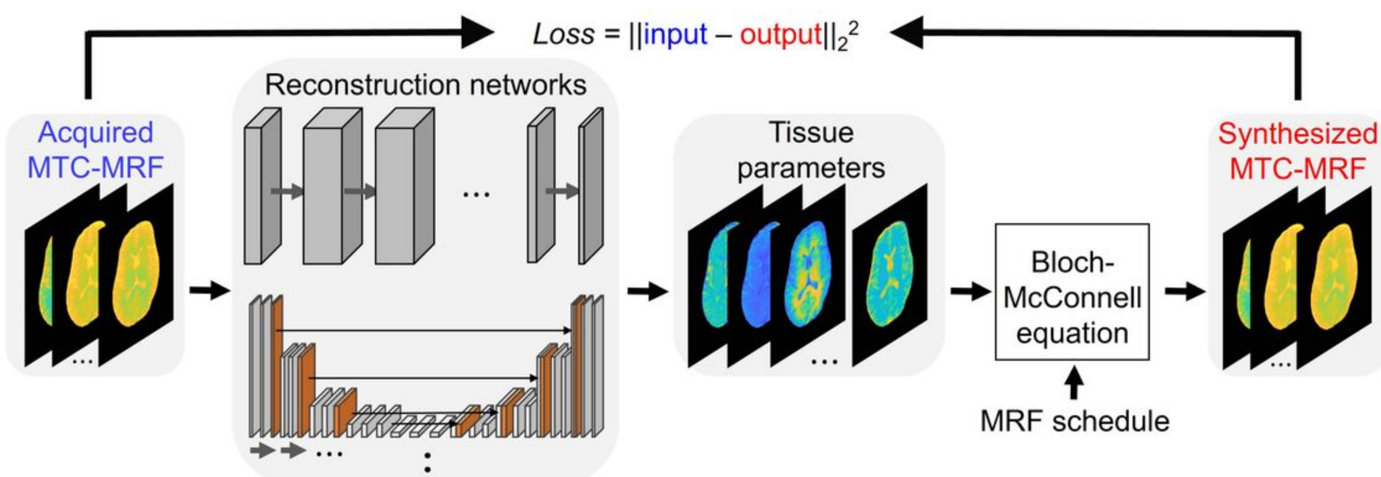

Figure 5

Fig. 5. Unsupervised-learning-based MRF reconstruction[40]. Label-free MRF images are fed into an eight-layer convolutional neural network (CNN) or U-Net as input and initial tissue parameter maps are generated. The Bloch McConnell analytical solution is then used to synthesize semisolid MT raw MRF images (output) based on the estimated tissue parameters. The output and input images are then used to calculate the squared error loss function. The architecture extracts key features from unlabeled input and finds a unique solution of the Bloch-McConnell equations by minimizing the loss function. At the end of the optimization procedure, quantitative water/saturation-transfer tissue parameter maps are extracted (see **Fig. 9**).

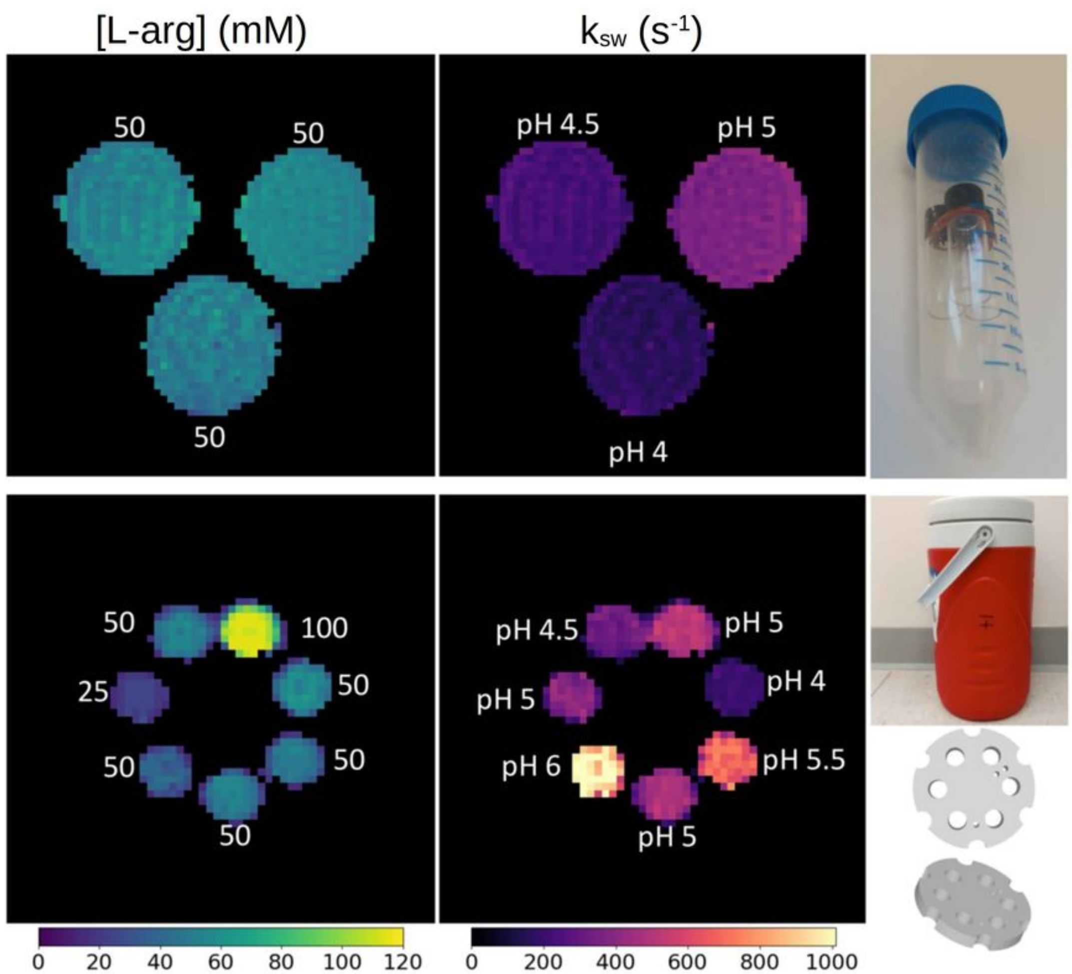

Figure 6

Fig. 6. Dot product-based parameter quantification example. **Top**. An L-arginine phantom was assembled, composed of three 2 mL vials placed inside a 50 mL Falcon tube (**right**). The phantom was scanned using a 9.4T Bruker scanner. **Bottom**. An L-arginine phantom composed of 15 mL vials was placed in a dedicated 3D holder inside a plastic water container (**right**). The phantom was scanned using a 3T clinical scanner. The white text atop the quantitative reconstructed maps represents ground truth values (**Left**).

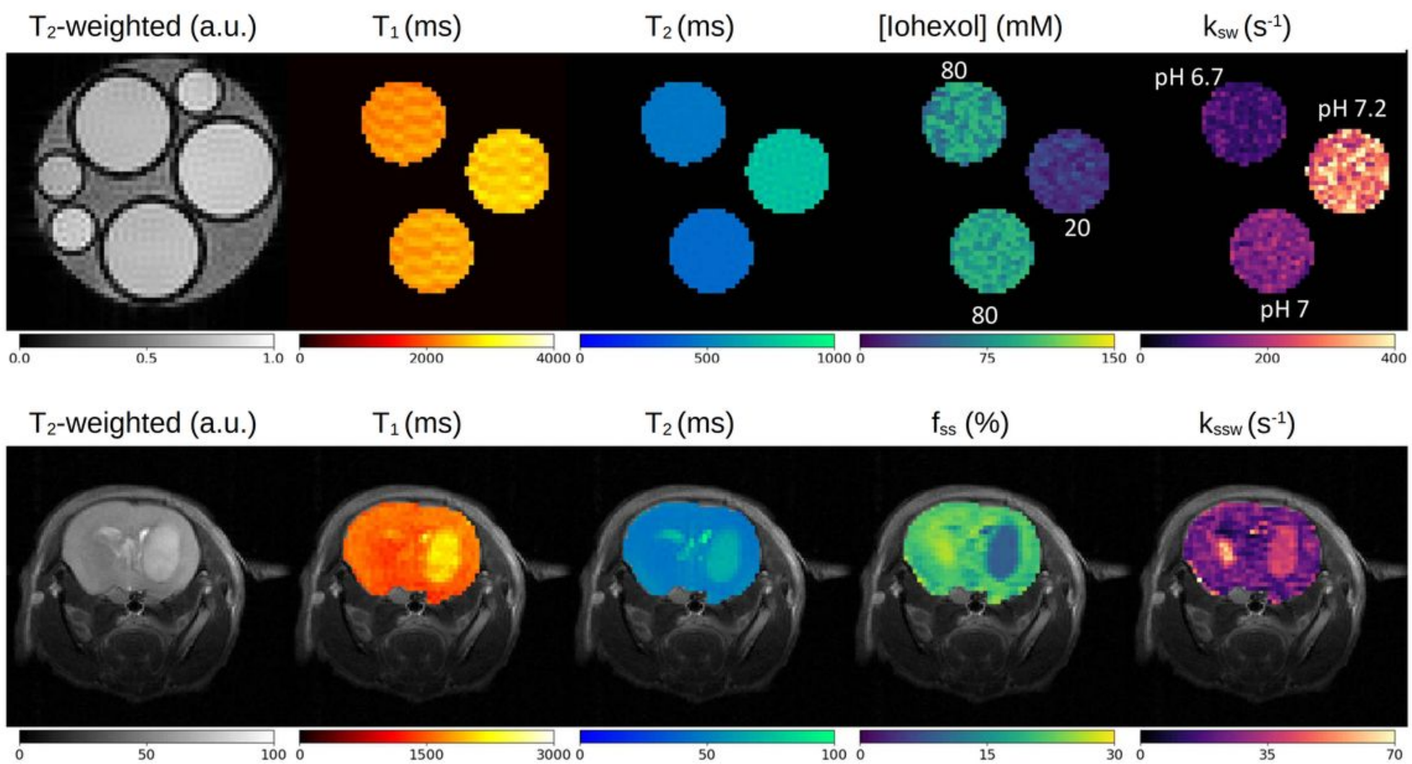

Figure 7

Fig. 7. Quantitative parameter maps reconstructed using sequential neural networks[29]. **Top**. Iohexol phantom imaged at 4.7T. The white text atop the quantitative reconstructed maps (**right**) represents ground truth values. **Bottom**. A tumor bearing mouse imaged at 7T.

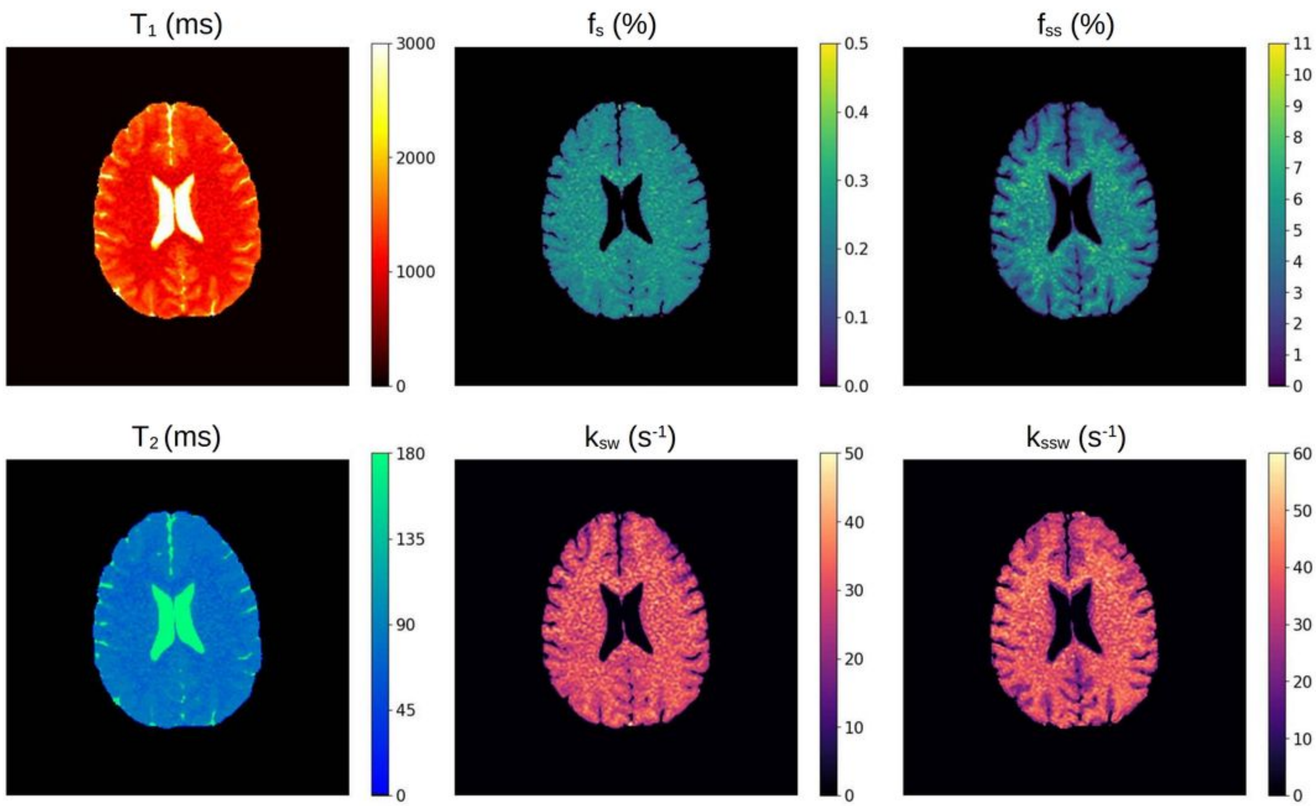

Figure 8

Fig. 8. Quantitative parameter maps from a healthy human volunteer scanned at 3T, obtained using a supervised and deep reconstruction network[41]. Each column represents a different proton pool: water (**left**), amide (**center**) and semisolid MT (**right**).

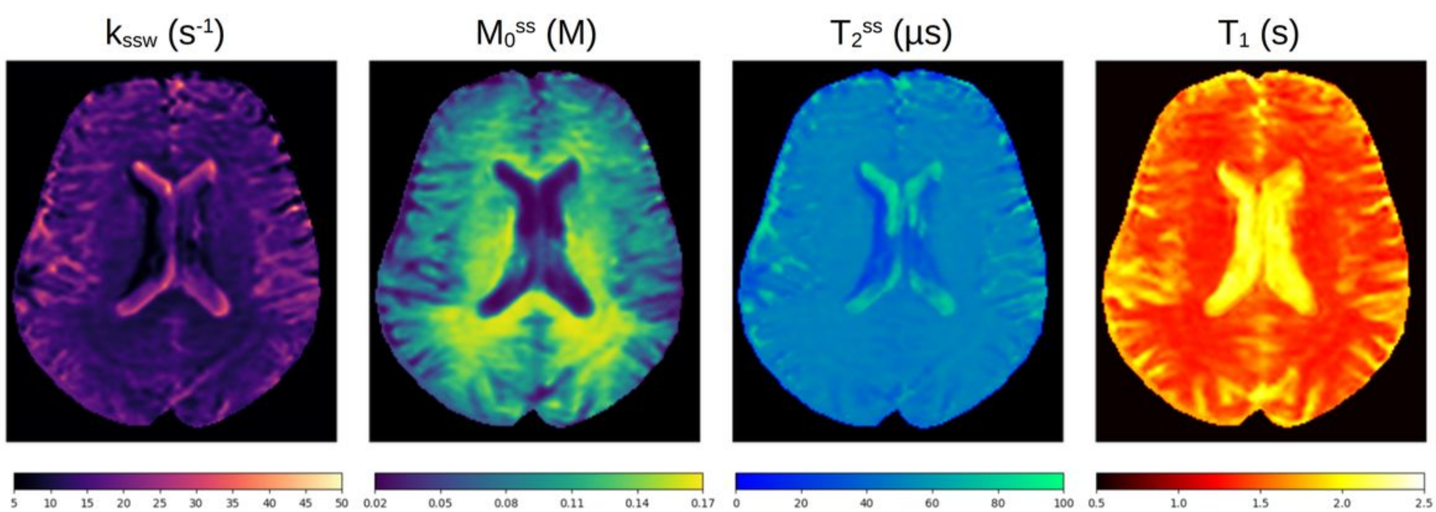

Figure 9

Fig. 9. Quantitative semisolid MT and water proton pool parameter maps from a healthy human volunteer scanned at 3T, obtained using an unsupervised deep learning approach[40].

Supplementary Files

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- tables.pdf